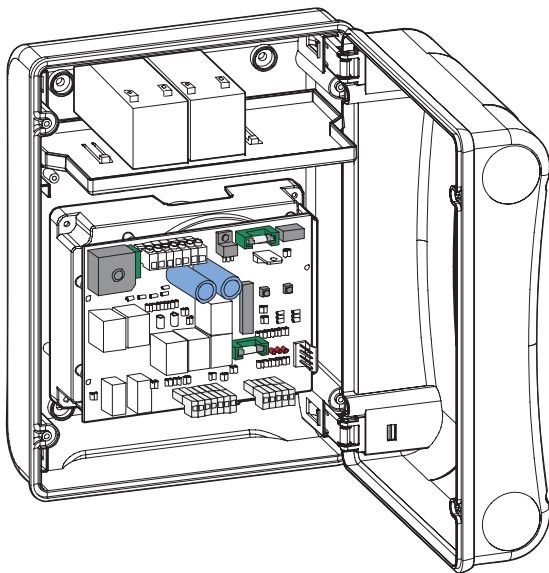


E012



FAAC

Leggere completamente questo manuale di istruzioni prima di iniziare l'installazione del prodotto.



Il simbolo evidenzia le note importanti per la sicurezza delle persone e l'integrità dell'automazione.



Il simbolo richiama l'attenzione sulle note riguardanti le caratteristiche od il funzionamento del prodotto.

Read this instruction manual to the letter before you begin to install the product.



Symbol highlights notes that are important for people's safety and for the good condition of the automated system.



Symbol draws your attention to the notes about the product's characteristics or operation.

Lire ce manuel d'instructions dans son entier avant de commencer l'installation du produit.



Le symbole met en évidence les remarques pour la sécurité des personnes et le parfait état de l'automatisme.



Le symbole attire l'attention sur les remarques concernant les caractéristiques ou le fonctionnement du produit.

Vor der Installation des Produkts sind die Anweisungen vollständig zu lesen.



Mit dem Symbol sind wichtige Anmerkungen für die Sicherheit der Personen und den störungsfreien Betrieb der Automation gekennzeichnet.



Mit dem Symbol wird auf Anmerkungen zu den Eigenschaften oder dem Betrieb des Produkts verwiesen.

Lean completamente este manual de instrucciones antes de empezar la instalación del producto.



El símbolo identifica notas importantes para la seguridad de las personas y para la integridad de la automatización.



El símbolo llama la atención sobre las notas relativas a las características o al funcionamiento del producto.

Lees deze instructiehandleiding helemaal door alvorens het product te installeren.



Het symbool is een aanduiding van opmerkingen die belangrijk zijn voor de veiligheid van personen en voor een goede automatische werking.



Het symbool vestigt de aandacht op opmerkingen over de eigenschappen of de werking van het product.

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EC DECLARATION OF CONFORMITY

Manufacturer : FAAC S.p.A.

Address: Via Benini, 1 - 40069 Zola Predosa BOLOGNA - ITALY

Declares that: E012 control board,

- conforms to the essential safety requirements of the following directives:
73/23/EEC and subsequent amendment 93/68/EEC.
89/336/EEC and subsequent amendment 92/31/EEC and 93/68/EEC

Additional note:

This product underwent tests in a typical uniform configuration
(all products manufactured by FAAC S.p.A.).

Bologna, 01 December 2006

The Managing Director

A. Bossi



1 INSTALLATION OF CONTROL UNIT

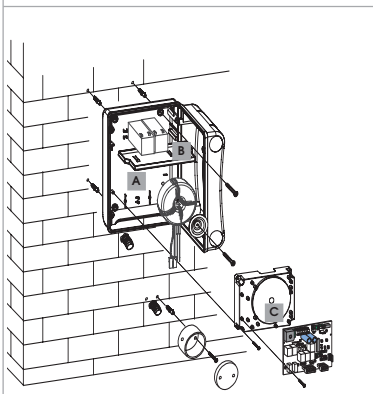
Procedure for securing components in the water-tight enclosure:

- 1- Secure the transformer to the water-tight enclosure in position (A)
- 2- If you are using buffer batteries (optionals), secure the relevant support in position (B).
- 3- Insert the batteries in the support.
- 4- Secure the control unit in position (C) with the 4 self-tapping screws (supplied), placing the supplied spacers between the control unit and the relevant support.
- 5- Secure the board support to the enclosure, using the supplied screws.

Using the four diam.6 plastic expansion plugs and relevant securing screws (not supplied), install the enclosure of the unit (to which all the accessories and pulse generators will then be connected), on the side of the gate supplied with 230V current (which must be connected to the toroidal transformer to change it to 13V) and near to the operator.

Route the electrical cables in the lower part of the enclosure, using adequate rigid tubes and/or hoses and appropriate connections.

Leave the cables inside the enclosure lengthy enough to enable you to wire up.



Secure two water-tight (IP 55) pull-boxes (not supplied), one for each operator, for wiring up.

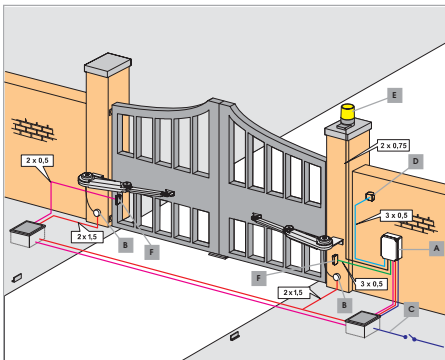
Use 12-terminal wiring bars (not supplied).



According to the mains voltage, the voltage coming out of the transformer can differ from the value given in paragraph 2.2.1. Before putting the control unit into service, we recommend you to check if the voltage coming out of the transformer is between 12 Vac and 14.5 Vac. This value is to be checked with load-free transformer, i.e. with the transformer powered but disconnected from the board.

2 ELECTRICAL CONNECTIONS

2.1 ELECTRICAL SYSTEM LAY-OUT



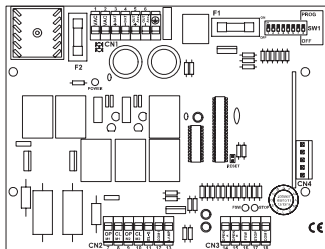
- A) Control board enclosure
- B) Pull boxes (not supplied)
- C) 3x1.5 mm²(2+T) 230V
- D) Key-operated selector switch OPTIONAL
- E) Flashing lamp 12 volt OPTIONAL
- F) Infrared photocells OPTIONAL

Notes:

- 1) To lay the electrical cables, use adequate rigid and/or flexible sheaths.
- 2) The figure shows an example of connection to the DOMOLINK operator. This connection layout can be used also for DOMOSWING.

2.2 DESCRIPTION OF CONTROL UNITS

2.2.1 TECHNICAL SPECIFICATIONS



SINGLE MULTI-FUNCTION CONTROL BOARD

Power supply voltage of the toroidal transformer: 230V - 50/60Hz
 Power supply voltage of the control unit: 13V - 50/60Hz
 Absorbed power: 7 W
 Motor max. load: 50 W (per motor)
 Accessories max. load: 12 Vdc 150mA
 Flashing light max. load: 12 Vdc 21W max.
 Operating temperature: -20°C / +55°C
 Protective fuse: 2
 Function logics: Automatic / Step-by-step automatic / Step-by-step semiautomatic.

Programming of functions:

Work time:

Pause time:

Leaf closing delay:

Adjustment of motor force:

Specifications of toroidal transformer 230V~:

Slow-downs:

Clutch device:

By Dip-Switch.

Self-learning during programming.

Self-learning during programming.

Two preset levels.

Two preset levels.

Primary 230V~ / sec. 13V~ / 150VA

At opening and closing during self-learning.

Electronic clutch adjustable on two preset levels.

2.2.2 PRESENTATION OF THE BOARD

DESCRIPTION OF TERMINALS

TERMINAL BOARD CN1 (power supply)

Terminals "1-2" (VAC-VAC): Connect, to these terminals, the secondary winding wires arriving from the toroidal transformer. Power ON is indicated by the lighting up of LED DL1 "power".

Terminals "3-4" (+BAT-BAT): Connect the 2 optional batteries to these terminals. During normal operation, the control unit keeps the batteries charged. The batteries come into operation when the transformer does not supply power.

During connection, respect the polarity indicated on the board.

A maximum of two batteries, connected in parallel, can be connected to the control unit.

Power supply provided by the batteries should be considered an emergency situation.

The number of possible manoeuvres depends on the quality of the batteries, the gate's structure (weight, dimensions, condition of the pivot, etc...), and on the time since power was cut, etc, etc.

Terminals "5-6" (+OUT-OUT): Output for 12Vdc power supply to the accessories.

The maximum load of the accessories must not exceed 150mA.

Terminal "⊕": Connect the earthing cable to this terminal. The earthing cable must be connected to the earthing system of the 230V~ power supply line.

The connection is absolutely necessary for correct operation of the control unit.

TERMINAL BOARD CN2 (motors)

Terminals "7-8" (OPM1-CLM1): The motor power cable must be connected to these terminals. For two-motor installations, the power cable of the motor which opens first must be connected to these terminals.

Terminals "9-10" (OPM2-CLM2): The power cable of the motor opening second must be connected to these terminals. For one motor only installation, no device must be connected to these terminals.

Terminals "11-12" (COM1-EL-S): (optional) The electric lock, if any, with 12 Vdc / 15W max power supply, must be connected to these terminals.

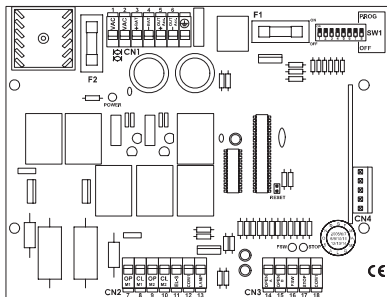
For 2-leaf installations, the electric lock must always be fitted on the leaf of motor M1, the one which opens first.

Terminals "12-13" (COM1-LAMP): Connect, to these terminals, the flashing light with power supply of 12Vdc 21W max. Before every manoeuvre, the flashing light pre-flashes for 0.5 sec.

Use a steady light flashing lamp; flashing is controlled by the control unit.

We advise you to install the flashing lamp before programming, because it indicates its phases.

TERMINAL-BOARD CN3 (commands and safety devices)



Terminals "14-18" (OPEN A-COM1): Normally open contact. Connect, to these terminals, any pulse generator (e.g. push-button, key selector, etc..) which, by closing a contact, generates a pulse for totally opening or closing the gate. The operation of this input is defined by dip-switch 3 - see paragraph 2.5.1.

Terminals "15-18" (OPEN B-COM1): Normally open contact. Connect, to these terminals, any pulse generator (e.g. push-button, key selector, etc..) which, by closing a contact, generates a pulse for partially opening the gate.

In two-motor installations (Dip-switch 5=ON); partial opening corresponds to the total opening of the leaf on which the motor M1 is installed.

In single-motor applications (Dip-switch 5=OFF), partial opening coincides with an opening of about ~80% of the total memory-stored opening.

If a total opening command (OPEN A) is given during the partial opening stage, the control unit memory stores the command, which is executed when partial opening has finished.

Terminals "16-18" (FSW-COM1): Normally closed contact. Connect, to these terminals, any safety device (photocell, pressure switch, safety edge etc..) which, by opening a contact, affects the gate's closing motion, reversing it to the mechanical opening stop, without disabling automatic re-closure if it is enabled. The status of this input is signalled by LED "FSW".

This contact can be disabled with dip-switch 6 - see paragraph 2.5.1.

If you decide to disable the photocells, you do not have to jumper the input.

The control unit is supplied with the photocell input disabled.

Terminals "17-18" (STOP-COM1): Normally closed contact. Connect, to these terminals, any pulse generator (push-button, key selector, etc..) which, by opening a contact, immediately stops the gate and disables all automatic functions. The status of this input is signalled by the "STOP"LED. The gate resumes its memory-stored cycle only by means of another total or partial opening pulse.

This contact can be disabled with dip-switch 7 - see paragraph 2.5.1.

If you decide to disable the stop command, you do not have to jumper the input.

The control unit is supplied with the stop command input disabled.

TERMINAL BOARD CN4

5-terminal connector for coupling receiver.

DIP-SWITCH

DP1(SW1): Parameter adjustment, see paragraph 2.5.1.

FUSES

F1: 16A T 250V 5X20.

F2: 10A T 250V 5X20.

LEDS	ON	OFF
POWER:	Control unit fed by transformer	No power supplied, or control unit fed by batteries.
FSW:	Safety device disengaged or input disabled.	Safety device engaged, contact open.
STOP:	Command inactive or input disabled.	Command active, contact open.

Indicated in bold: status of LEDs with the gate closed, and control unit fed by transformer.

2.2.3 POWER SUPPLY CONNECTION

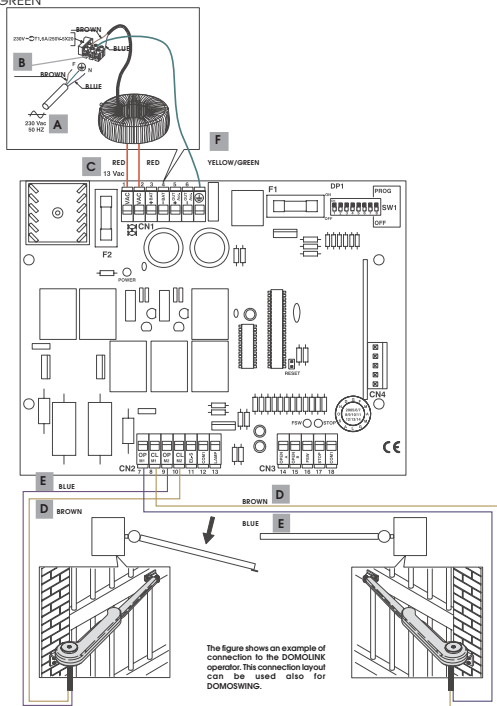
As concerns power supply:

1 Use a 3 x 15 mm2 cable for conveying the 230V mains current, protected by a fuse or by a 10 Amp circuit breaker, routing from the dwelling to the gate.

The figure shows an example of connection to the DOMOLINK operator. This connection layout can be used also for DOMOSWING.

2nd CASE: RIGHT LEAF OPENS FIRST TOWARD THE INSIDE

- A) 230 Volt Mains power supply coming from the electrical panel of the dwelling.
Protected by a differential switch or by a 10 Amp fuse.
3 x 1.5 mm² line from the dwelling to the pilaster
- B) The terminal with fuse, inserted in the enclosure, is used for connecting the 230 V~ mains supply to the primary winding of the toroidal transformer.
- C) 12 Vdc power supply from toroidal transformer.
- D) BROWN
- E) BLUE
- F) YELLOW/GREEN

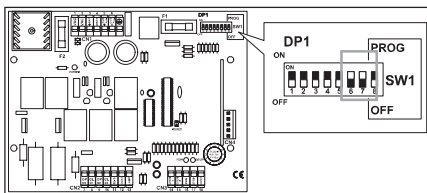


2.3.1 AN ESSENTIAL CONTROL

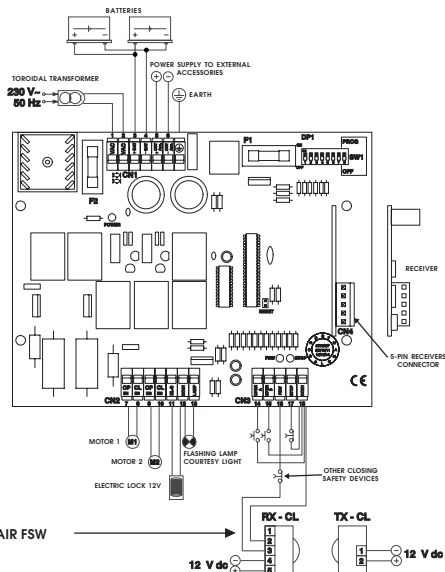
If you are not using safety devices (photocell, pressure switch, edge, etc.) or pulse generators (push-button, key selector, etc.), make sure that cursors 6 and 7 on the dip-switch are disabled (position "ON"). If you decide to disable these commands, you do not have to jumper the input.

This control is essential, otherwise the motor will not start.

IMPORTANT: The dip-switch cursors are shown in white in the following drawings.



2.4 WIRING OF (OPTIONAL) ACCESSORIES



2.4.1 CONNECTION OF (OPTIONAL) BATTERIES

Terminals "+ BAT - BAT". Connect the 2 optional batteries to terminals "+ BAT - BAT".

During normal operation, the control unit keeps the batteries charged.

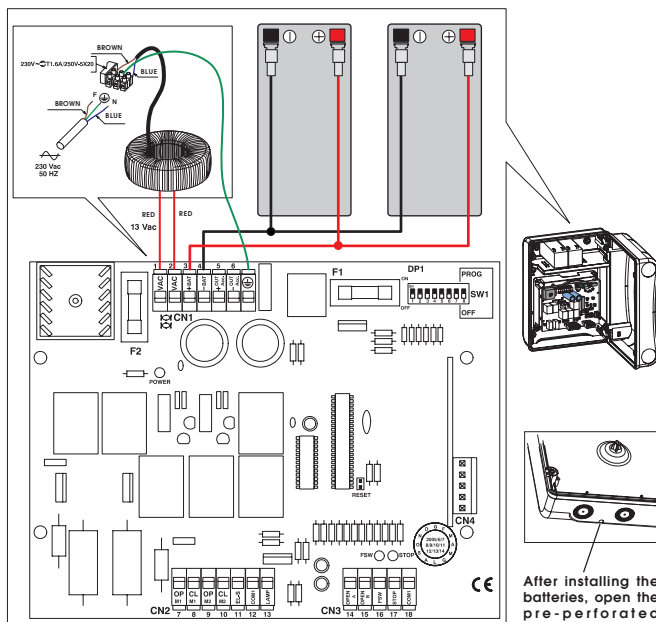
The batteries come into operation when the transformer does not supply power.

During connection, respect the polarity indicated on the board.

A maximum of two 12 Vdc - 1.2 Ah batteries, connected in parallel, can be connected to the control unit.

Power supply provided by the batteries should be considered an emergency situation.

The number of possible manoeuvres depends on the quality of the batteries, the gate's structure (weight, dimensions, condition of the pivot, etc.), and on the time since power was cut, etc.



2.5 PROGRAMMING

2.5.1 PROGRAMMING THE OPERATING PARAMETERS

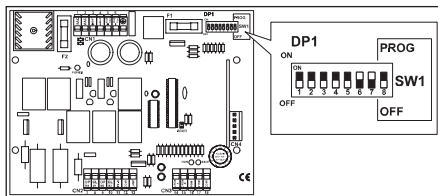
The multi-function control board has a dip-switch (DP1) for programming the operating parameters. Dip-switch DP1 is used for setting various parameters, such as the operating logic, automatic closing, and enabling/disabling the safety devices, etc..

All selectable parameters are shown in the table below.

To avoid compromising the condition and operation of the control unit, parameters must be set with the control unit powered down.

We advise you to select the parameters you require before programming.

DIP-SWITCH	FUNCTION	ON	OFF
1	Electronic clutch	Max. force / Min. sensitivity	Min. force / Max. sensitivity
2	Automatic closing	Enabled	Disabled
3	Total opening function	Opens - Stop - Closes - Stop	Opens- Closes (without Stop)
4	Leaf closing delay	3-sec. delay	1.5 sec. delay
5	Number of motors	Two-motor operation	Single motor operation
6	Photocells input	Disabled	Enabled
7	Stop input	Disabled	Enabled
8	Programming	During programming	Normal operation



2.5.2 ELECTRONIC CLUTCH

This control unit has an electronic clutch device based on the control of the current absorbed by each motor. This device is very important for the purpose of safety.

It is active during both closing and opening. When it operates, it reverses gate movement without disabling automatic closing if enabled.

If it operates for two consecutive times, the control unit goes into "STOP" status, disabling all automatic functions. This is because, when the clutch operates twice consecutively, it means that the obstacle is still present and it could be dangerous to perform any other manoeuvre. When the obstacle has been removed, give a "START" pulse and the control unit will resume the memory-stored cycle.

If the clutch operates three consecutive times, and for more than 120 seconds, the control unit performs an "EMERGENCY" procedure: i.e. it executes opening until the opening mechanical stop is reached, and then closes, if automatic closing has been enabled. In this way the control unit synchronises itself automatically, and automatically re-calculates the opening and closing stop points.

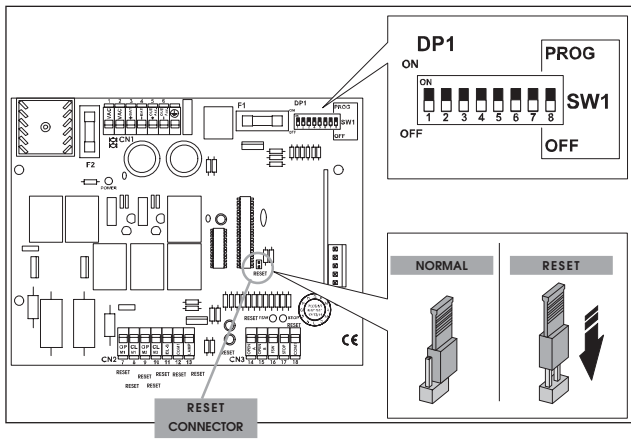
The "EMERGENCY" procedure occurs while the motors are turning at low speed.

2.5.3 PROGRAMMING OPENING, CLOSING TIMES AND GATE PAUSE

During this stage, the control unit memory stores work time and any pause time.

To access programming, follow the instructions below:

- 1- Release the operators (see the instructions on the mechanical operators) and take the leaves to about halfway of the opening travel.
- 2- Relock the operators (see the instructions on the mechanical operators) and power up the system.
- 3- Turn ON dip-switch 8 on the control unit. The flashing lamp lights up on steady beam to indicate entry to the programming procedure.
- 4- Give a total opening pulse with the "START" push-button or with the radio control (if already memory stored) and the leaves will begin to move.
The first manoeuvre the control unit carries out is closing.
- 5- Make sure that both leaves perform the closing manoeuvre and that the leaf with motor M2 moves first.
If this does not happen, stop gate movement with a "RESET" pulse, jumper connecting the two "RESET" pins to each other and using the connector (see figure).
- 6- Cut the power supply and take the connector back to its normal position (see figure).
- 7- Make the suitable corrections to the motor connections, and repeat from point 4, changing over the cables of the motors on connector CN2 (see paragraph 2.3.).
- 8- When they reach the closing mechanical stop point, the leaves pause for two seconds, and then begin the opening stage
- 9- When both leaves reach the opening mechanical stop point, the pause time count begins.
- 10- When the required pause time has elapsed, give a total opening pulse.
The leaves begin the closing stage.
- 11- When the leaves reach the closing stop, programming has finished.
- 12- Turn OFF dip-switch 8 - the flashing lamp goes OFF to finish programming.



2.5.4 OPERATING LOGICS

AUTOMATIC LOGIC (DIP-SWITCH 2=ON / DIP-SWITCH 3=OFF)				
GATE STATUS	PULSES			
	OPEN A	OPEN B	STOP ⁽¹⁾	PHOTOCELLS ⁽²⁾
Closed	Opens leaves and re-closes after pause time	Partially opens the gate ⁽³⁾	Disables OPEN commands	No effect
Open in pause	Closes at once	Closes at once	Stops pause time count	Stops operation and on release, if pause time exceeded, closes after 5 sec
Closing	Reverses gate movement	No effect	Stops operation	Reverses motion
Opening	No effect	No effect	Stops operation	No effect
STEP-BY-STEP AUTOMATIC LOGIC (DIP-SWITCH 2=ON / DIP-SWITCH 3=ON)				
GATE STATUS	PULSES			
	OPEN A	OPEN B	STOP ⁽¹⁾	PHOTOCELLS ⁽²⁾
Closed	Opens leaves and re-closes after pause time	Partially opens the gate ⁽³⁾	Disables OPEN commands	No effect
Open in pause	Closes at once	Closes at once	Stops pause time count	Stops operation and on release, if pause time exceeded, closes after 5 sec.
Closing	Stops gate motion and, at next pulse, opens	No effect	Stops operation	Reverses motion
Opening	Stops gate motion and, at next pulse, closes	No effect	Stops operation	No effect
SEMI-AUTOMATIC LOGIC (DIP-SWITCH 2=OFF / DIP-SWITCH 3=OFF)				
GATE STATUS	PULSES			
	OPEN A	OPEN B	STOP ⁽¹⁾	PHOTOCELLS ⁽²⁾
Closed	Opens leaves	Partially opens the gate ⁽³⁾	Disables OPEN commands	No effect
Open	Closes at once	Closes at once	Stops operation	On release, closes after 5 sec.
Closing	Reverses gate movement	No effect	Stops operation	Reverses motion
Opening	Reverses gate movement	No effect	Stops operation	No effect
SEMI-AUTOMATIC STEP-BY-STEP LOGIC (DIP-SWITCH 2=OFF / DIP-SWITCH 3=ON)				
GATE STATUS	PULSES			
	OPEN A	OPEN B	STOP ⁽¹⁾	PHOTOCELLS ⁽²⁾
Closed	Opens leaves	Partially opens the gate ⁽³⁾	Disables OPEN commands	No effect
Open	Closes at once	Closes at once	Stops operation	On release, closes after 5 sec.
Closing	Stops gate movement and, at next pulse, opens	No effect	Stops operation	Reverses motion
Opening	Stops gate movement and, at next pulse, closes	No effect	Stops operation	No effect

⁽¹⁾ The operation described in the tables is obtained if the STOP input is enabled (dip-switch 6 = OFF). When the input is disabled, the command has no effect.

⁽²⁾ The operation described in the tables is obtained if the input of the photocells is enabled (dip-switch 7 = OFF). When the input is disabled, the command has no effect.

⁽³⁾ In single motor installations, partial opening of the gate corresponds to about 60% of the memory stored opening. In two-motor installations, partial opening corresponds to the total opening of the leaf on which motor M1 is installed.

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